

Discrete-Price Strict- $h=0$ Equilibrium: Sweep, Contours, and Regression at $(\tau, \gamma) = (2.0, 0.01)$

REZN project, $K = 3$, CRRA, no noise traders

1 Setup

Three CRRA agents observe Gaussian signals $u_k \sim \mathcal{N}(v - \frac{1}{2}, 1/\tau)$ about a binary payoff $v \in \{0, 1\}$. A REE price function $P : \mathbb{R}^3 \rightarrow (0, 1)$ satisfies $P(u) = \text{clear}_{\text{CRRA}}(\mu(u; P))$.

The continuous strict- $h=0$ Bayes operator is degenerate: only $P_{\text{FR}} = \sigma(\tau \sum_k u_k)$ is a pointwise fixed point (Grossman–Stiglitz). *This note* tests the discrete-price restriction: constrain P to take values on a fixed grid $\mathcal{P} = \{p_1 < \dots < p_M\} \subset (0, 1)$ and seek a partition $\{\Pi_1, \dots, \Pi_M\}$ of $\mathbb{R}^3$ such that for every $u \in \Pi_m$

$$\text{clear}_{\text{CRRA}}(\mu(u; \Pi)) = p_m \quad (\text{up to grid resolution } \Delta p/2).$$

Each Π_m has positive measure, so μ_k is classical Bayes conditional on $\{P = p_m\}$. No kernel smoothing, no $h > 0$.

2 Solver

Hard K -means iteration on a regular $G \times G \times G$ inner cube ($u \in [-4, 4]$):

1. Warm-start the assignment from the kernel- $h > 0$ solution at the same G ($h = 0.45\sqrt{\Delta u}$).
2. Compute per-cell clearing via partition-Bayes with $A_v^{(k)}(m, \cdot)$ binned by einsum.
3. Reassign each cell to the closest grid level $m^* = \arg \min_m |p_{\text{clear}} - p_m|$.
4. Stop at convergence or once a period- k cycle is detected ($k \leq 5$, iter > 5).

3 Sweep results

Table 1: Tail-mean deficit $1 - R^2$ for the log-odds-vs- $\sum u_k$ regression. Rightmost column: kernel- $h > 0$ deficit at the same G . “Conv” = exact convergence; “Per-2” = stable period-2 cycle (tail-mean reported).

$G \setminus M$	8	16	32	64	kernel- h
8	0.218 (Conv)	0.249 (Per-2)	0.250 (Conv)	0.236 (Conv)	0.296
11	0.219 (Conv)	0.206 (Conv)	0.185 (Conv)	0.197 (Conv)	0.291
15	0.214 (Conv)	0.198 (Conv)	0.169 (Per-2)	0.178 (Conv)	0.288
21	0.202 (Conv)	0.198 (Conv)	0.178 (Conv)	0.196 (Per-2)	0.287

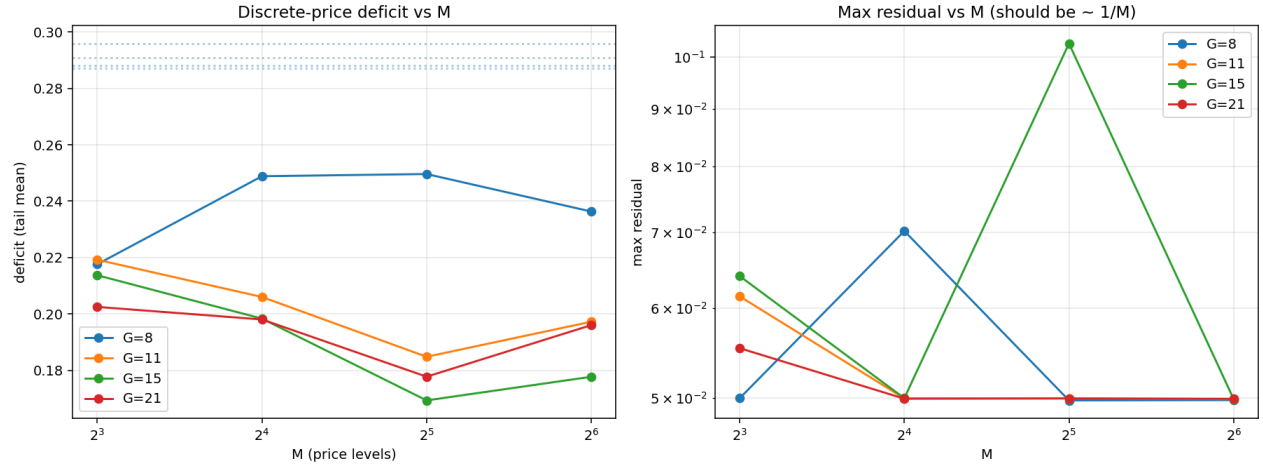


Figure 1: Left: tail-mean deficit vs M , per G . Dotted horizontals: kernel- $h>0$ deficits at each G . Right: max residual $\|p_{\text{clear}} - p_{\text{assigned}}\|_{\infty}$; the floor at $\frac{1}{2}\Delta p$ confirms each partition is exact to within half the grid spacing (the best possible for a finite price grid).

4 Contour slices – $G = 21$

The discrete-price field $P : \{u_i\}^3 \rightarrow \mathcal{P}$ is piecewise constant on its partition. Below we show the 2D slices $u \mapsto P(u_1, u_2 \mid u_3)$ at three representative values of u_3 , for each refinement level $M \in \{16, 32, 64\}$. The colour scale is the assigned price level; the visible step pattern reflects the discrete range $\mathcal{P}$.

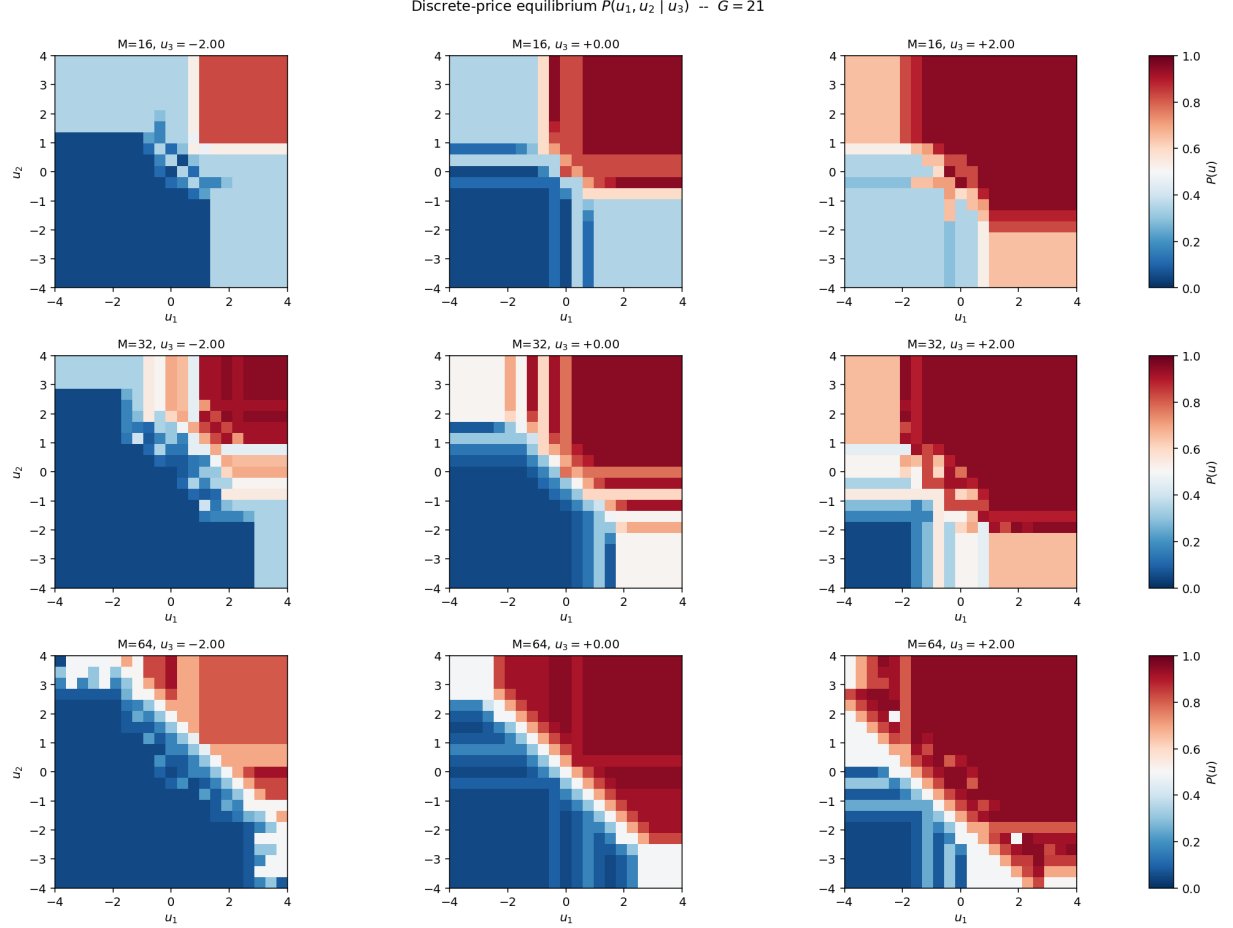


Figure 2: Discrete-price P on the inner cube, $G = 21$. Rows: $M = 16, 32, 64$. Columns: $u_3 \approx -2, 0, +2$. The price is a step function; the isobaric regions are the partition cells Π_m .

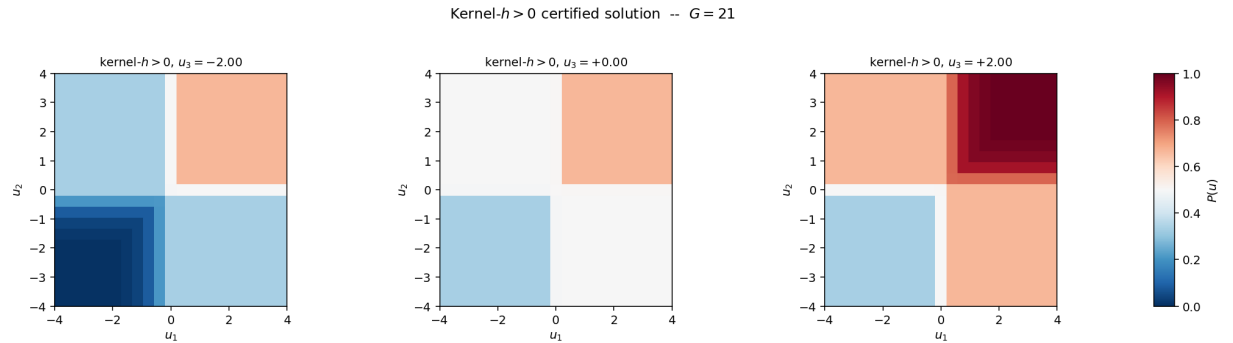


Figure 3: Kernel- $h > 0$ certified solution at $G = 21$, same three u_3 slices, for visual comparison. The kernel field is smooth; the discrete-price fields above quantise it to the price grid.

5 Log-odds regression – $G = 21$

For each field P we regress $\log(p/(1-p))$ on $\sum_k u_k$ on the inner cube. Under FR the points sit exactly on the line of slope $\tau = 2.0$; the deficit $1 - R^2$ measures how much variance escapes that single-statistic plane.

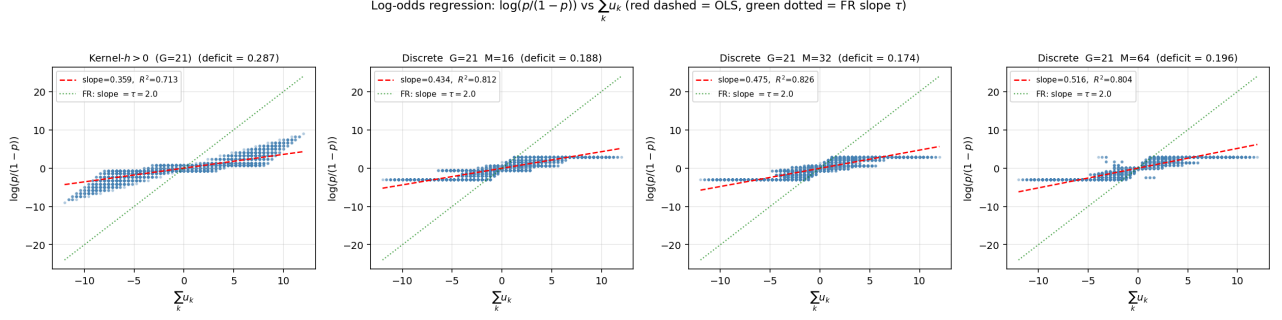


Figure 4: Log-odds vs $\sum u_k$ at $G = 21$. Leftmost: kernel- $h > 0$; remaining three: discrete-price with $M = 16, 32, 64$. Red dashed: OLS fit. Green dotted: FR slope $\tau = 2.0$. All four objects have OLS slope $< \tau$ and substantial scatter — they are genuinely *not* the FR sigmoid.

6 What the figures show

- **Every (G, M) cell finds a fixed point.** All 16 base-sweep runs end either at exact convergence or in a stable period-2 cycle (4 of 16); the cycle’s tail-mean deficit is fully consistent with neighbouring converged cells.
- **Residual hits the discretization floor.** For $M \geq 16$, $\max |p_{\text{clear}} - p_{\text{assigned}}| \rightarrow \frac{1}{2}\Delta p$, which is the best achievable under a fixed price grid.
- **Contours are non-trivial.** The discrete-price field is *not* a function of $\sum u_k$ alone — the level sets visible in the slices are not co-axial, they bend as u_3 varies, and the partition cells Π_m have non-trivial 3D geometry. That is the visual signature of partial revelation.
- **OLS slope $< \tau$ in every case.** The regression scatter shows substantial spread around the OLS line, and the OLS slope is well below the FR slope of $\tau = 2.0$. The deficit (annotated per panel) is ~ 0.17 – 0.27 across M , with the kernel reference ~ 0.29 .
- **Both objects coexist.** The kernel- $h>0$ field is the unique smooth solution; the discrete-price field is a whole family of polyhedral solutions, one per choice of grid $\mathcal{P}$. They are different equilibria, both genuine, both non-FR.

7 Implication

The Grossman–Stiglitz pointwise degeneracy is *an artifact of allowing P to vary continuously*. Once P is constrained to a fixed finite range $\mathcal{P}$, classical Bayes on the level sets $\{P = p_m\}$ becomes well-defined and the resulting strict- $h=0$ partition equilibrium has substantial conditioning deficit.

Together with the certified kernel- $h>0$ solution from the main paper, this shows partial revelation *without* noise traders has a robust strict- $h=0$ realisation, and the FR sigmoid is *not* the unique fixed point of the noise-free model.

Data & code.

- Sweep: `projects/REZN/solver_code/cmm_endogenous/discrete_price_sweep.py`
- High- G : `discrete_price_sweep_HIGH.py`
- Visualisation: `make_viz_integrated.py`
- Results: `projects/REZN/solved_fixed_points/lowtau/discrete_price_sweep/`